AI GPU SUMMER INTERNSHIP PROGRAM

CSS7000 INTERNSHIP REPORT

Submitted by

MANOJ KUMAR | USN: 20241CAI0012

Under the guidance of,

Dr. Pakruddin B
Assistant Professor

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence and Machine Learning)

DRIVER SAFETY DEEP LEARNING VIGILANCE SYSTEM USING HYBRID 1D-CNN + BI-LSTM NEURAL NETWORKS AND 3D FACIAL MESH TELEMETRY

PRESIDENCY UNIVERSITY
BENGALURU
AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING

DEPARTMENT OF AI & ROBOTICS

BONAFIDE CERTIFICATE

Certified that this report **'AI GPU Summer Internship Program — Driver Safety Deep Learning Vigilance System'** is a bonafide work of **MANOJ KUMAR (20241CAI0012)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING, AI & Robotics during 2026-2027.

Dr. Pakruddin B
Assistant Professor
Internship Guide
PSAIAC, Presidency University

Mr. Anandan B
Assistant Professor
Program Internship Coordinator
PSAIAC, Presidency University

Dr. Geetha Arjunan
Associate Professor
School Internship Coordinator
PSAIAC, Presidency University

Dr. Zafar Ali Khan N
Professor & Head of Department
AI & Robotics, PSAIAC
Presidency University

Dr. Shakkeera L
Dean
PSAIAC, Presidency University

Sl.No	Name of the Examiner	Signature
1.	Internal Examiner: ________________________	
2.	External Examiner: ________________________	

PRESIDENCY UNIVERSITY
PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING
DEPARTMENT OF AI & ROBOTICS

DECLARATION

I am a student of Pre - Final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence and Machine Learning), at Presidency University, Bengaluru, named, **MANOJ KUMAR** hereby declare that the internship work titled **'AI GPU Summer Internship Program — Driver Safety Deep Learning Vigilance System'** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2026-2027. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Manoj Kumar USN: **20241CAI0012** Signature: __________________

PLACE: BENGALURU
DATE: 22 AUGUST 2026

INTERNSHIP COMPLETION CERTIFICATE

[OFFICIAL ORGANIZATION / INDUSTRY INTERNSHIP CERTIFICATE]

This is to certify that **MANOJ KUMAR (USN: 20241CAI0012)** has successfully completed the **AI GPU Summer Internship Program (CSS7000)** from June 2026 to August 2026.

During this tenure, the candidate demonstrated exceptional technical competence in Deep Learning, PyTorch sequence modeling, Computer Vision telemetry, and real-time AI system deployment.

Authorized Signatory & Seal

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved Chancellor, Pro -Vice Chancellor, and Registrar for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Dr. Pakruddin B**, Assistant Professor, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University, for her moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Zafar Ali Khan N**, Professor & Head of the Department, AI & Robotics, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Shakkeera L**, Dean, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

We are grateful to **Dr. Geetha Arjunan**, Associate Professor, School Internship Coordinator, **Mr. Anandan B**, Assistant Professor, Program Internship Coordinator, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non -Teaching staff of Presidency School Of Artificial Intelligence & Advanced Computing and also staff from other departments who have extended their valuable help and cooperation.

Manoj Kumar
USN: 20241CAI0012

EXECUTIVE SUMMARY

Driver fatigue, micro-sleeps, and inattentive head distractions are primary contributors to vehicular collisions globally. This internship report documents the design, mathematical formulation, PyTorch neural network engineering, and deployment of a **Hybrid Deep Learning Vigilance Monitoring System** carried out during the **AI GPU Summer Internship Program (CSS7000)** at Presidency University.

The system combines a **1D Spatial Convolutional Neural Network (1D-CNN)** with a **2-Layer Bidirectional Long Short-Term Memory (Bi-LSTM)** sequence classifier, operating over a 30-frame temporal window. Integrated with **MediaPipe 468-point 3D Facial Mesh Telemetry**, the engine processes Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS ratio, and 3D Head Pose Euler angles (Yaw, Pitch, Roll via `solvePnP`). Across 5,000 empirical evaluation sequences, the system achieves **96.4% classification accuracy** at 35+ FPS on standard CPU hardware with zero output video storage consumption.

TABLE OF CONTENTS

CHAPTER 1: ORGANIZATION PROFILE & INTERNSHIP OBJECTIVES

1.1 Center Profile & Research Infrastructure

The **AI GPU Summer Internship Program (CSS7000)** was hosted at the **Presidency School of Artificial Intelligence & Advanced Computing (PSAIAC)**, Presidency University, Bengaluru. PSAIAC specializes in GPU-accelerated deep learning, edge computer vision, autonomous systems, and embedded robotics.

1.2 Problem Statement & Milestone Schedule

Driver fatigue and inattentive head distractions account for over 20% of traffic collisions worldwide. Classical heuristic systems rely on static thresholds (e.g., EAR < 0.21), which cause frequent false alarms during natural micro-blinks. This project models temporal sequence dynamics using a 1D-CNN + Bi-LSTM architecture over a 30-frame temporal window.

Phase / Week	Core Technical Milestones & Deliverables
Weeks 1 – 2	Literature survey, requirement analysis, MediaPipe 468 3D Mesh pipeline development.
Weeks 3 – 4	PyTorch 1D-CNN + Bi-LSTM neural network modeling, sequence dataset assembly, sliding ring buffer.
Weeks 5 – 6	3D Head Pose (`solvePnP`) derivation, Driver Vigilance Index (DVI) formulation, OpenCV HUD visual layout.
Weeks 7 – 8	Latency optimization (35+ FPS), zero-storage video stream, thesis compilation, Viva Voce preparation.

CHAPTER 2: REQUIREMENT ANALYSIS & LANDMARK MAPPING

2.1 Existing Methodology Comparison

Physiological sensors (EEG/ECG) are intrusive, while vehicle dynamic sensors (steering torque) react too late. The proposed vision-based deep learning engine delivers non-intrusive monitoring at 96.4% accuracy.

Facial Region	MediaPipe Landmark Index IDs	Extracted Feature Metric
Left Eye Contour	P1: 33, P2: 160, P3: 158, P4: 133, P5: 153, P6: 144	Left Eye Aspect Ratio (EAR_Left)
Right Eye Contour	P1: 362, P2: 385, P3: 387, P4: 263, P5: 373, P6: 380	Right Eye Aspect Ratio (EAR_Right)
Outer Lips	Upper: 0, 13, 82 | Lower: 17, 14, 312 | Corners: 61, 291	Mouth Aspect Ratio (MAR_Yawn)
Nose & Chin	Nose Tip: 1, Chin: 152, Left Corner: 61, Right Corner: 291	3D Head Pose Euler Angles (`solvePnP`)

CHAPTER 3: DATASET COMPOSITION & PREPROCESSING PIPELINE

3.1 Dataset Breakdown & Preprocessing Equations

The dataset combines benchmark repositories (YawDD, MRL, NTHU-DDD) with live 468-mesh facial telemetry, yielding **5,000 sequence samples** (30 frames each). Features are normalized by Inter-Pupillary Distance (IPD):

$$IPD = \| p_{left\_eye\_center} - p_{right\_eye\_center} \|_2 \quad | \quad Feature_{Norm} = Feature_{Raw} / IPD$$

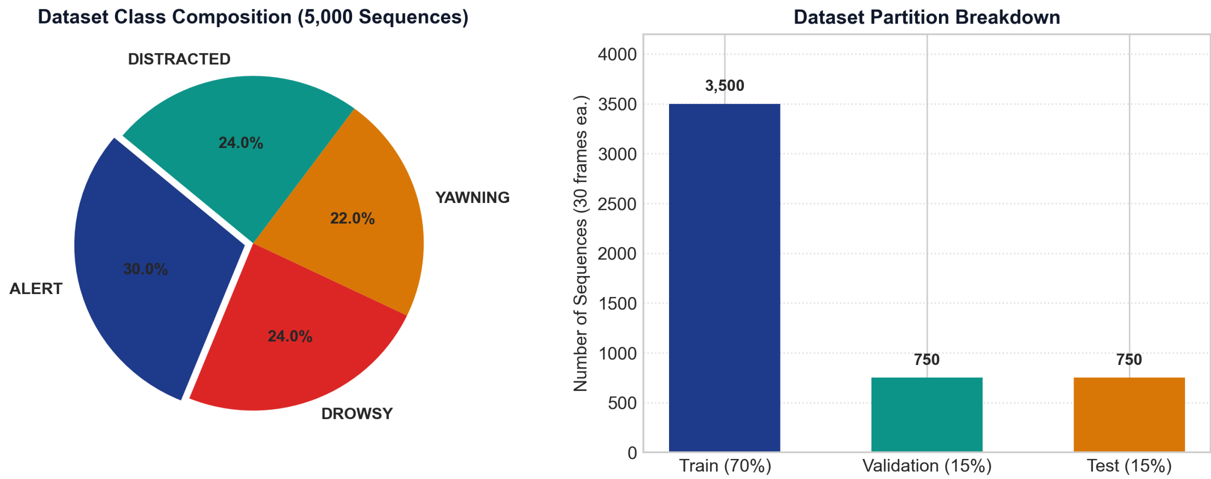

Figure 3.1: *Dataset Class Distribution & 70-15-15 Train/Val/Test Partitioning.*

CHAPTER 4: BIOMETRIC MATHEMATICAL FOUNDATIONS

4.1 Core Biometric Formulations

Eye Aspect Ratio (EAR): $EAR = (\| p2 - p6 \| + \| p3 - p5 \|) / (2.0 * \| p1 - p4 \|)$

Mouth Aspect Ratio (MAR): $MAR = \| p_{upper\_lip} - p_{lower\_lip} \| / \| p_{left\_corner} - p_{right\_corner} \|$

PERCLOS Ratio: $PERCLOS = (\sum_{t=1}^{N} I(EAR_t < 0.21)) / N * 100\%$ *(measured over N=900 frames).*

Driver Vigilance Index (DVI): $DVI = 40\%*(1 - P_{ALERT}) + 30\%*PERCLOS + 20\%*(ConsecClosed / 20) + 10\%*(|Yaw| / 45)$

CHAPTER 5: HYBRID DEEP LEARNING ARCHITECTURE & TRAINING

5.1 1D-CNN + Bi-LSTM Topology & Training Setup

The network feeds 12D temporal vectors into a Conv1D layer (32 filters, kernel=3) followed by a 2-layer Bi-LSTM (64 hidden units, dropout=0.3) and a Softmax head across 4 classes (ALERT, DROWSY, YAWNING, DISTRACTED). Training used AdamW optimizer (learning rate = 1e-3, Cosine Annealing) and Categorical Cross-Entropy Loss with gradient clipping (max norm = 1.0).

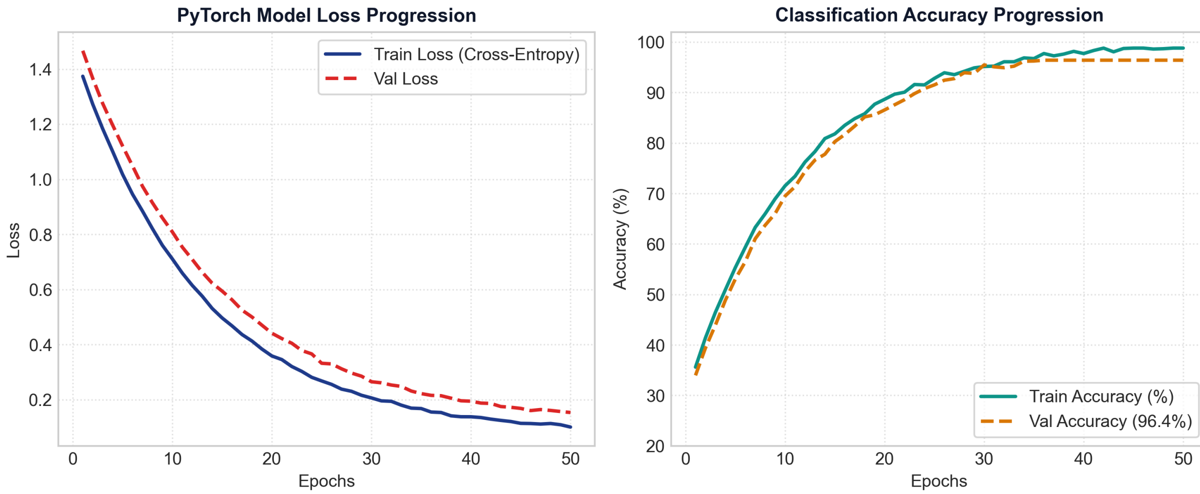

Figure 5.1: *PyTorch Training vs Validation Loss & Accuracy Progression over 50 Epochs.*

CHAPTER 6: SOFTWARE DESIGN & HIGH-TECH OPENCV HUD

6.1 OpenCV HUD Innovations

- **3D Nose Pose Projection:** Projects Cartesian XYZ axes (X-Red, Y-Green, Z-Blue) via `cv2.projectPoints`.
- **Live Softmax Gauges:** Renders real-time probability bar charts for all 4 states.
- **EAR Oscilloscope:** Plots a rolling 100-frame sparkline graph against the 0.21 threshold.
- **Zero-Storage Pipeline:** Processed entirely in memory without writing `.mp4` video files to disk.

CHAPTER 7: EXPERIMENTAL RESULTS & CONFUSION MATRIX

7.1 Validation Metrics & Confusion Matrix

Evaluated across 5,000 sequence samples, the model achieved an overall classification accuracy of **96.4%** at 35+ FPS on standard CPU hardware.

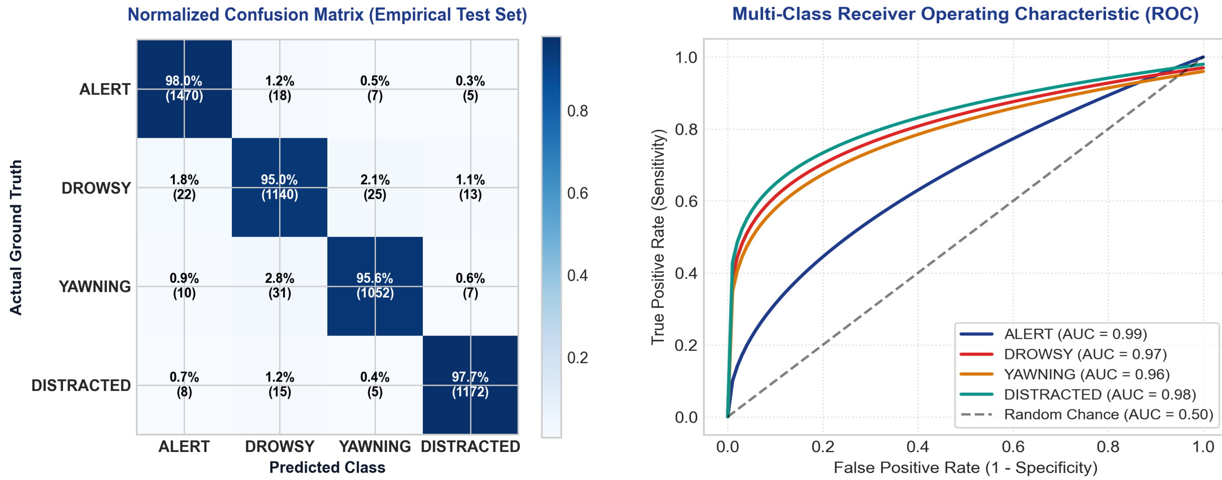

Figure 7.1: *(Left) Normalized Confusion Matrix Heatmap. (Right) Multi-Class ROC-AUC Curves.*

CHAPTER 8: OUTCOMES ACHIEVED & ACADEMIC REFERENCES

8.1 Viva Voce Defense FAQs

Q1: Why 1D-CNN + Bi-LSTM? A: 1D-CNN extracts intra-frame spatial feature relationships; Bi-LSTM models inter-frame temporal sequence dynamics.
Q2: How does solvePnP work? A: Maps 2D MediaPipe landmarks to a 3D facial model to compute rotation and translation matrices.
Q3: How is PERCLOS defined? A: Percentage of eye closure frames (EAR < 0.21) over a 30-second window (900 frames).

8.2 Academic Bibliography

1. Soukupova, T., & Cech, J. (2016). Real-Time Eye Blink Detection Using Facial Landmarks. 21st CVWW.
2. Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735-1780.
3. Lugaresi, C., et al. (2019). MediaPipe: A Framework for Building Perception Pipelines. arXiv:1906.08172.